PROJECT 3 - ZOMATO DATA ANALYSIS

1. Introduction

This project performs an end-to-end exploratory data analysis (EDA) on the Zomato restaurant dataset to understand customer preferences, pricing patterns, ratings behavior, and restaurant distribution across cities and cuisines.

The objective is to clean the dataset, analyze key features, visualize patterns, and generate business insights useful for restaurant platforms and food businesses.

2. Data Cleaning & Preparation

The dataset contained missing values and inconsistent data types.
The following steps were performed:

• Removed missing values using dropna()
• Converted price range and cost columns to numeric format
• Created cost categories (Budget, Mid-range, Expensive, Luxury)
• Created rating categories (Poor, Average, Good, Excellent)
• Cleaned dataset exported for reproducibility

This ensures accurate and consistent analysis.

3. Exploratory Data Analysis (EDA)

The following analyses were performed:

Restaurant distribution

Examined how restaurants are distributed across:

- Cost categories
- Cities
- Cuisines

Customer behavior analysis

Studied:

- Rating distribution
- Impact of online delivery
- Impact of table booking
- Votes vs rating relationship

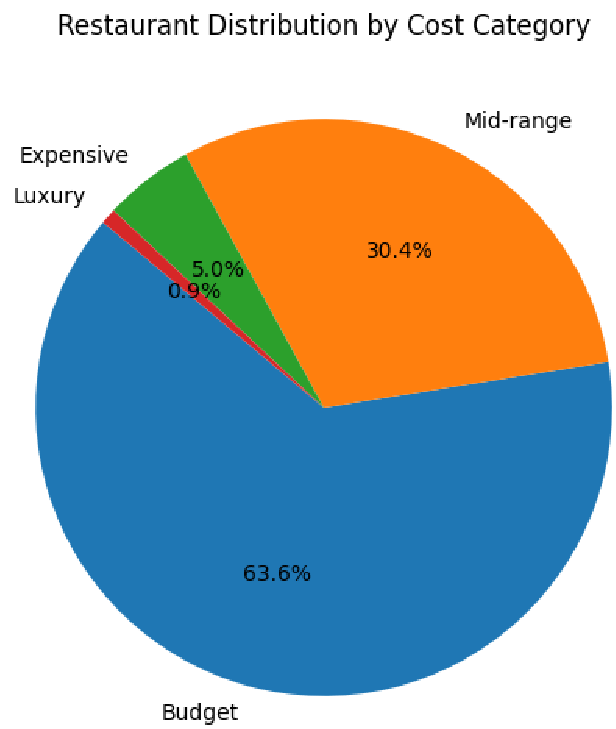
Market insights

Identified:

- Most popular cuisines
- Cities with highest restaurant concentration
- Highest rated restaurants

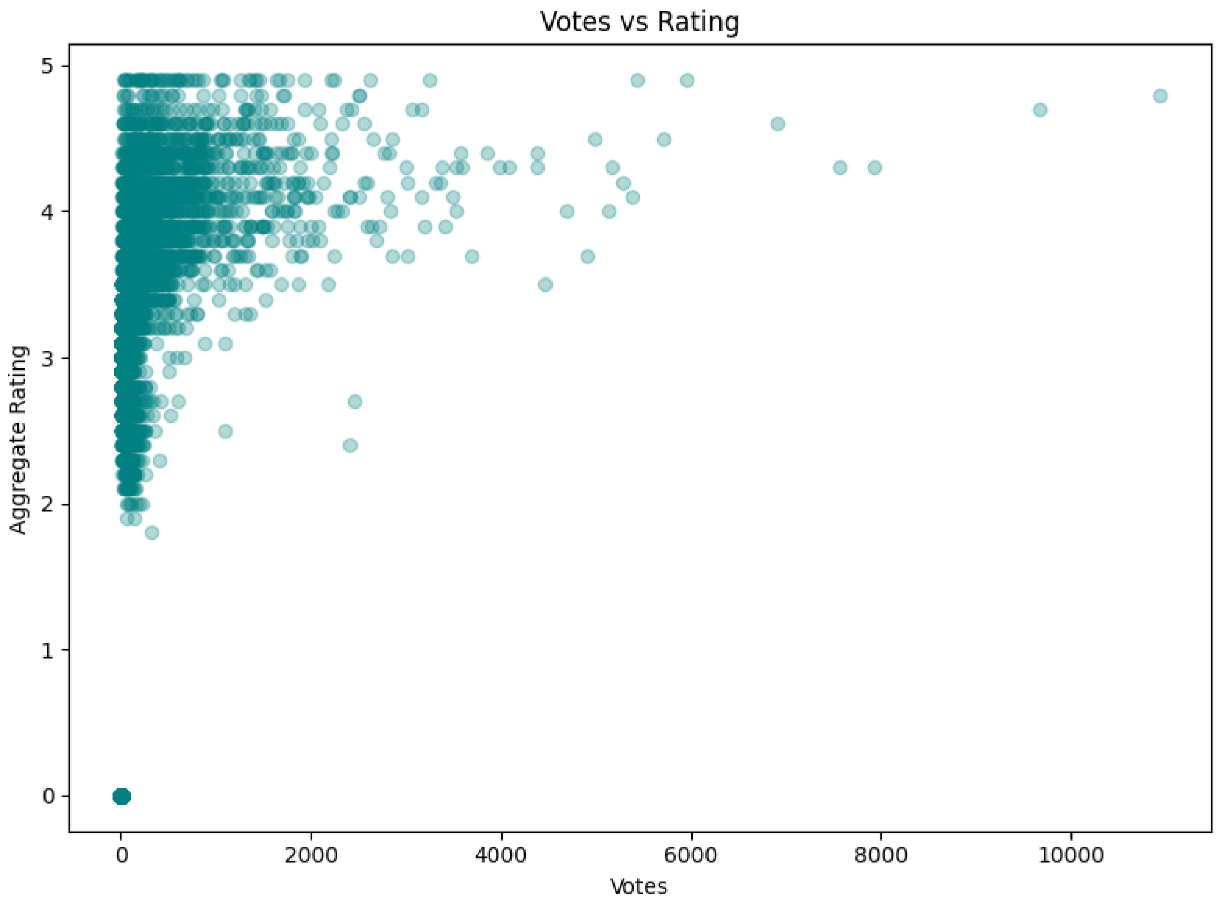
4. Key Visualizations

Visualization 1: Cost Category Distribution



This chart shows most restaurants fall under budget and mid-range pricing, indicating a price-sensitive customer base.

Visualization 2: Votes vs Rating Relationship



This chart shows a positive relationship between number of votes and ratings. Restaurants with more customer engagement tend to have higher ratings.

5. Three Actionable Business Insights ⭐ (IMPORTANT)

Insight 1: Budget restaurants dominate the market

Most restaurants fall into the budget and mid-range category, showing high demand for affordable dining options.
Businesses should focus on competitive pricing and value meals.

Insight 2: Online delivery improves ratings

Restaurants offering online delivery have higher average ratings than those without delivery.
Investing in delivery services can improve customer satisfaction and visibility.

Insight 3: Popular cuisines drive demand

North Indian, Chinese, and Fast Food are the most common cuisines across cities.
New restaurants should consider these cuisines for higher demand and market success.

6. Conclusion

This project demonstrates a complete data analysis workflow including data cleaning, exploratory analysis, visualization, and business insight generation.

The findings highlight pricing trends, customer preferences, and key factors influencing restaurant success. These insights can help food platforms and restaurant owners make data-driven decisions.